Verification supplement to Parabola cap sets over $\mathbb{F}_{6561}$: two order-41 coefficient cosets of opposite quadratic character

Abstract

This supplement records an alternate finite upper certificate for the order-41 subgroup, its 24 obstruction representatives, and the correspondence between the mathematical argument and the portable verification package. It also describes a direct check of the attaining five-coefficient construction and an independent arithmetic check of the opposite-character coset obstruction. The alternate upper certificate and the complete-family certificate use the same discriminant hypergraph.

1 Conventions and finite inputs

Throughout, a cap set means a subset of an $\mathbb{F}_3$-vector space containing no three distinct points with sum zero. Put

$$F = \mathbb{F}_3[z]/(z^8 + z^2 + 2), \qquad h = z^6 + z^5 + 2z^2 + z, \qquad H = \langle h \rangle, \qquad g = z^3 + z^2 + 2, \qquad D = H \cup gH.$$

The element g is primitive and $h = g^{160}$ has order 41. Field elements are encoded by integers

$$\sum_{i=0}^{7} a_i 3^i \longleftrightarrow \sum_{i=0}^{7} a_i z^i, \qquad a_i \in \{0, 1, 2\}.$$

Thus h has identifier 993. This convention specifies the generator used in every table below. Exponents are residues modulo 41 represented by integers from 0 to 40. For $T \subseteq \mathbb{Z}/41\mathbb{Z}$, write

$$S(T) = \{h^i : i \in T\}, \qquad P(S) = \{(x, cx^2) : x \in F^*,\ c \in S\}.$$

The coefficient-discriminant method appears in Grace and Voloch [2, Lemma 2.4]; the exact specialization needed here, including repeated coefficients, is proved in the main paper.

Let $\mathcal{G}$ be the 3-uniform hypergraph whose edges are the distinct triples $\{i, j, k\}$ for which

$$\Delta(h^i, h^j, h^k) = -(h^{i+j} + h^{i+k} + h^{j+k})$$

is a square in F, with zero included as a square. Then $P(S(T))$ is a cap set exactly when T is independent in $\mathcal{G}$. There are 10,660 triples, 5,740 edges, and 4,920 nonedges; none of these discriminants is zero. The zero convention is retained in the definition and in verification. Translations $i \mapsto i + t$ and the eight multipliers

$$M = \langle 3 \rangle = \{1, 3, 9, 27, 40, 38, 32, 14\}$$

preserve $\mathcal{G}$. No other multiplier symmetry is assumed.

2 The opposite-character coset check

The main paper proves that a cap coefficient set in D cannot meet both cosets. Here is the exact finite correspondence checked independently of the H-certificate. For $a \in H$ and $b \in gH$, write $r = b/a$. The two repeated-label discriminants are

$$\Delta(a, a, b) = a^2(r - 1), \qquad \Delta(a, b, b) = -a^2 r(r - 1).$$

The ratio $-r$ is a nonsquare, so exactly one discriminant is a square. Since no two elements of D are antipodal, the normalized equation from the main paper has nonzero leading coefficient and neither excluded endpoint is a root. Hence the square discriminant produces three distinct zero-sum points. This uses the converse under the nonantipodal hypothesis proved in the main paper; the published [2, Lemma 2.4] states only necessity.

The independent checker `verification/verify_two_cosets.py` rebuilds field arithmetic from the polynomial convention, proves that g runs through all 6560 nonzero elements, and checks that H consists of squares while gH consists of nonsquares. It checks both repetition patterns for every unordered coefficient pair. All $2\binom{41}{2} = 1640$ same-coset pairs are safe; all $41^2 = 1681$ cross-coset pairs are forbidden in exactly one repetition pattern. As a concrete witness, the coefficient multiset with identifiers $(1, 1, 38)$ has normalized ratio 1151 and produces the distinct points with identifiers

$$(1151, 1927), \quad (1, 1), \quad (2046, 1342),$$

whose two coordinates separately sum to zero in F.

For an additional relation check, the program enumerates all $\binom{82}{3} = 88\,560$ distinct-label triples. The numbers classified as forbidden by their discriminant, by the number of labels in gH, are 5740, 16 851, 16 769, and 5740, respectively; none has zero discriminant. Those mixed-triple counts are diagnostic rather than an input to the maximum or classification, because every mixed set already contains a forbidden pair. Scaling the H result by g gives exactly 328 attaining sets in each coset, or 656 in D; this is two displayed classes, not a claim of one orbit preserving D.

3 The zero link and an alternate upper certificate

Let L be the graph on $\{1, \ldots, 40\}$ in which $\{i, j\}$ is an edge if $\{0, i, j\}$ is an edge of $\mathcal{G}$. It has 420 edges. Every $\mathcal{G}$-independent set containing 0 gives an L-independent set after 0 is removed. Exact enumeration gives

Cardinality	0	1	2	3	4	5
Number of independent sets	1	40	360	960	808	192

There are 2,361 independent sets in total and none of size greater than 5. The counts of maximal independent sets of sizes 3, 4, and 5 are 24, 144, and 192, respectively. Here "maximal" means maximal under inclusion.

For clarity, the complete enumeration can be checked by a simple recurrence. Start with the empty set A and the ordered candidate list $1, \ldots, 40$. Record A. For each candidate v, recurse with $A \cup \{v\}$ and retain only candidates greater than v and nonadjacent to v. By induction, the retained candidates are already nonadjacent to all of A. The recurrence therefore visits only independent sets. Conversely, every independent set has a unique increasing sequence of vertices, so it occurs on exactly one branch. The recorded distribution is thus a statement of exhaustive enumeration, not just a list of valid examples.

The 192 independent five-sets of L form 24 orbits under multiplication by M. Every orbit has size eight. Indeed, a nontrivial stabilizer in this cyclic group of order eight would contain multiplication by -1. That involution has no fixed point on $\{1, \ldots, 40\}$; it cannot preserve a set of odd cardinality.

Proposition 1. *No independent subset of $\mathcal{G}$ has more than five vertices.*

Proof. Translate a hypothetical independent six-set to contain 0. Removing 0 leaves an independent five-set of L. Table 1 lists one representative from each of its 24 multiplier orbits. Each representative contains the displayed edge of $\mathcal{G}$. Multiplication by M preserves edges, so every member of every orbit contains an edge as well. This contradicts independence. Larger independent sets would contain independent six-subsets. □

The fact that all displayed representatives contain 1 is an output of the orbit enumeration. It is not a normalization imposed on the original 41-vertex problem. In particular, the argument never pins a second coefficient using an unproved multiplier action.

Table 1: All 24 multiplier orbits of independent five-sets in the zero link. The third column is an internal edge of $\mathcal{G}$. Each orbit has eight members.

	Representative	Forbidden triple
1	$\{1,4,5,6,24\}$	$\{1,4,6\}$
2	$\{1,4,5,9,24\}$	$\{1,5,9\}$
3	$\{1,4,6,24,34\}$	$\{1,4,6\}$
4	$\{1,4,9,24,34\}$	$\{1,4,34\}$
5	$\{1,5,6,8,24\}$	$\{1,5,8\}$
6	$\{1,5,6,18,24\}$	$\{1,5,18\}$
7	$\{1,5,8,9,24\}$	$\{1,5,8\}$
8	$\{1,5,9,22,38\}$	$\{1,5,9\}$
9	$\{1,5,9,24,38\}$	$\{1,5,9\}$
10	$\{1,5,9,37,38\}$	$\{1,5,9\}$
11	$\{1,5,18,24,38\}$	$\{1,5,18\}$
12	$\{1,6,8,24,34\}$	$\{1,6,8\}$
13	$\{1,7,15,27,32\}$	$\{1,7,27\}$
14	$\{1,7,27,32,33\}$	$\{1,7,27\}$
15	$\{1,7,27,33,35\}$	$\{1,7,27\}$
16	$\{1,7,27,35,37\}$	$\{1,7,27\}$
17	$\{1,8,9,24,34\}$	$\{1,8,24\}$
18	$\{1,8,20,24,34\}$	$\{1,8,20\}$
19	$\{1,8,20,30,34\}$	$\{1,8,20\}$
20	$\{1,9,10,37,38\}$	$\{1,9,37\}$
21	$\{1,12,15,18,38\}$	$\{1,12,18\}$
22	$\{1,12,22,33,35\}$	$\{1,12,33\}$
23	$\{1,12,22,33,38\}$	$\{1,12,33\}$
24	$\{1,15,18,24,38\}$	$\{1,15,24\}$

4 Direct verification of attainment

For the fixed generator h, let

$$T_0 = \{0, 2, 6, 13, 23\}, \qquad S_0 = \{1, 1063, 3121, 5401, 4873\}$$

where the second set is written in field identifiers. In addition to checking that T_0 contains no edge, the verifier checks the normalized equations directly. For each coefficient multiset $\{a, b, c\}$ from S_0, using a fixed order of its entries, it evaluates

$$(a + c)t^2 + 2ct + (b + c), \qquad t \in F \setminus \{0, -1\}.$$

There are $\binom{7}{3} = 35$ multisets: five with all entries equal, 20 with exactly two equal, and ten with three distinct entries. Thus there are $35 \cdot 6559 = 229{,}565$ evaluations. The only roots are the five all-equal cases at $t = 1$, each representing the same point three times. It suffices to use a single ordering of each multiset: any alleged zero-sum triple can be reordered to that ordering and normalized by its nonzero second first-coordinate. Repeated entries cause no exception to this argument.

The five punctured parabolas are disjoint, since equality of (x, ax^2) and (y, by^2) with $x, y \neq 0$ implies $x = y$ and then $a = b$. Consequently the attaining cap set has $5(6561 - 1) = 32{,}800$ points.

5 The complete-family certificate

The main proof uses a second enumeration: all independent subsets of $\mathcal{G}$ containing 0. The supplied family $\mathcal{C}$ has 929 members, with cardinality distribution

$$(|\mathcal{C}_1|, |\mathcal{C}_2|, |\mathcal{C}_3|, |\mathcal{C}_4|, |\mathcal{C}_5|) = (1, 40, 360, 488, 40).$$

Every member is checked for validity. For each increasing tuple $A \in \mathcal{C}$ and every integer $v > \max A$, the verifier checks whether $A \cup \{v\}$ is independent; if it is, that extension must also appear in $\mathcal{C}$. There are 10,182 such extension obligations and 928 compatible extensions. Starting with $\{0\}$, induction over increasing prefixes proves that every normalized independent set appears. In particular, absence of six-sets is meaningful only together with this closure check.

The 40 five-sets through 0 yield $41 \cdot 40/5 = 328$ five-sets in all by translation and incidence counting. The main paper proves that the action $i \mapsto 3^j i + t$ has trivial stabilizer on any independent five-set. The verifier additionally expands the 328 images of T_0, checks that they are distinct and independent, and compares their normalized members with the 40 certified five-sets through 0.

6 Reproduction and scope of the checks

The Parallel Attack computational companion, version `companions-v1`, contains five mathematical data files and two verifiers requiring only Python 3.9 or later and its standard library. Its README gives the commands and the role of each check. Both `verification/verify.py` and `verification/verify_two_cosets.py` are required for the two-coset classification: the latter uses the H classification checked by the former. Paths are resolved relative to each program. Each result is written only to a new output path chosen by the caller.

The field model is checked before the certificates. With the stated polynomial arithmetic, successive powers of $g = z^3 + z^2 + 2$ (identifier 38) enumerate all 6,560 nonzero residues and then return to 1. Thus every nonzero element of the quotient ring is a unit, which verifies that the quotient is a field. The equality $h = g^{160}$ and the order of h are also checked. Every hypergraph row is then recomputed from its field discriminant, rather than accepted from the stored edge flag. The direct root check uses coefficient convolution and polynomial reduction as an additional arithmetic implementation.

The finite checks establish the exhaustive facts used in the proofs. Their interpretation depends on the normalized-equation correspondence proved in the main paper. The two upper certificates share the same hypergraph, so agreement between them does not constitute two unrelated mathematical premises. The two-coset theorem adds a symbolic pair obstruction and imports the H classification after an explicit scaling

check; it does not require a new 82-vertex independence search. Neither certificate asserts an unrestricted coefficient maximum, an ambient maximum cap size, or completeness of the attaining cap set.

The construction and predicate antecedents are credited to [2] and its associated code repository [1]. All numerical assertions in this supplement can be checked from the supplied finite data and programs.

References

[1] Cassie Grace. capsets: code accompanying algebraic capsets. `https://github.com/cassie-grace/capsets`, 2026. Revision `d74441822bcf`, 5 February 2026.

[2] Cassie Grace and José Felipe Voloch. Algebraic capsets. *Journal of Combinatorial Designs*, 34(10):459–463, 2026. Publisher version. `https://doi.org/10.1002/jcd.70024`.